

UG-C-3002-BCA

2022-26

Full Marks : 70

Time : 3 hours

Answer from both the Groups as directed.

The figures in the right-hand margin indicate marks.

Candidates are required to give their answers in their own words as far as practicable.

GROUP—A

Answer any *four* questions. 10 × 4

- 1. What is the need of JVM in execution of any Java program ? Discuss the structure of any Java program.**
- ② Define loop. Explain all looping statements available in Java with suitable example.**
- 3. What is meaning of access control of a variable and function in Java ? Explain the different access control mechanism in Java with the help of a suitable program.**

(Turn Over)

4. Define method overloading. Write a program to calculate the area of different shapes namely *Circle*, *Rectangle* and *Triangle* using the concept of method overloading.
5. Explain the different usage of final keyword with suitable example.
6. Explain how a string class object can be created using an existing '**StringBuffer**' class. Also explain how can you find the location of the last occurrence of '*a*' in the string "**Java Programming**".
7. What is multithreaded programming ? Explain how threads are created in Java. Explain the need of thread synchronization with an example.
8. What is an exception ? Write an exception subclass which throws an exception if the variable *age* passed as argument to a method and the value of age is less than 18.

(3)

GROUP-B

Answer *all* the questions. 3×10

9. Explain any two benefits of inheritance.
10. What is a global variable ?
11. What are the advantages of java programming over C++.
12. Give one example of type casting in java.
13. What are the characteristics of static members in Java ?
14. Define constructor overloading.
15. Write a java program to display even numbers between 0 to 20.
16. Explain any two usage of super keyword.
17. Define wrapper class in java.
18. What do you mean by Abstract class ?

2023(New)*Time : 3 hours**Full Marks : 70*

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Group – A

Answer any **four** questions of the following :

10×4 = 40

1. Define data types. Write the different data types available in Java.
2. What is loop ? Also write its types with example.
3. What is constructor overloading ? Explain it with suitable example.
4. Define inheritance. Explain multilevel inheritance with example.

5. Differentiate between abstract class and interface. Explain how multiple inheritance can be achieved through interface.
6. Define package. Write down the steps to create the package. Also, write the program to use the newly created package.
7. What is Multithreading ? Write the life cycle of a thread.
8. Define exception and its types. Write a program to create user defined exception.

Group – B

Answer all questions :

10×3 = 30

9. What is the difference between C++ and Java ?
10. Define Java Virtual Machine (JVM).
11. What is type casting ?
12. Write the usage of ? : operator.
13. Define static member.
14. What is the use of wrapper class ?

15. List any five string functions.
16. Why is java called a platform independent language ?
17. Explain each word in the statement `public static void main (String args [])`.
18. What is the use of final keyword ?



2025

Time : 3 hours

Full Marks : 70

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The figures in the margin indicate full marks.

Answer from both the Groups as directed.

Group – A

Answer any four questions of the following :

10×4 = 40

1. Discuss how Java differs from C and C++ in terms of memory management and platform independence. Discuss the role of Data Types in defining variables and their impact on program efficiency.
- ② Explain the basic features of object oriented programming language. Explain the functionality of Java Virtual machine.

use abstract keyword
no variable: constant

3. Discuss different types of loops and their applications. Provide an example to show how loops can simplify repetitive tasks in a program.
4. What is an abstract class ? Explain the use of abstract class with an example.
5. What are interfaces ? Illustrate how interfaces can be used for implementing multiple inheritance.
6. Explain the life cycle of a thread. Write a program that creates two threads one thread displays numbers from 10 to 1 and the other thread displays numbers from 10 to 1.
7. Define exception. Which keywords are used for exception handling ? Write a program that catches negative exception (user defined exception). This is caused when a negative number is entered by a user.
8. What is string class ? How is it different from StringBuffer class ? Write a program to count number of vowels in a string.

Group – B

9. Answer all questions of the following :

3×10 = 30

- (a) Java is platform independent language ?
justify.
- (b) What is the use of super keyword ?
- (c) What is finalization in Java ?
- (d) What is package ? How are package created and accessed in java ?
- (e) What is demon thread ?
- (f) What is the major difference between array and vector ?
- (g) Why thread synchronization important for multithreading process ?
- (h) Differentiate between equals () method and == operator.
- (i) Explain any three string functions used in JAVA.
- (j) Explain each component of the statement "public static void main (String args[])".



2026

Full Marks : 70

Time : 3 hours

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indicate marks.*

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GROUP—A

Answer any *four* questions from the
following :

10 × 4

1. Describe Java as an Object oriented programming language. What are the key features of Java ?
2. Explain JVM, JDK and JRE with suitable diagrams and explain all functionalities.

3. What is constructor in Java ? Explain the different types of constructors. Does Java, provide default constructor ? Explain your answer with suitable example.
4. How is method overloading different from method overriding ? Explain with the help of a suitable code.
5. Define an Interface. How does it differ from an Abstract class ? Write a Java program to create an interface Shape with the getArea() method. Create three classes Rectangle, Circle, and Traingle that implement the Shape interface. Implement the getArea() method for each of the three classes.
6. Differentiate between Thread class and Runnable interface. Write a program that creates two threads; one thread displays odd numbers between 1 and 20, and the other thread displays even numbers between 1 and 20.

(3)

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Madhusu
Newton

Answer
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7. Define exception. Which keywords are used for exception handling ? Write a program to create custom exception called InvalidAgeException. If a user enters an age less than 18, the program will throw this exception.
8. Differentiate among String, StringBuilder and StringBuffer in Java ? Explain any five string function with an example.

GROUP—B

Answer *all* questions :

3 × 10

9. Discuss all three usages of final keyword.
10. What is the use of this and super keyword ?
11. Describe garbage collection in Java.
12. Why we need both run and start method in multithreading ?

13. What is the role of stack in exception handling?

14. How can you set a priority of a thread?

15. What is the purpose of static block, static variable and static method in Java?

16. What is package? How are package created and accessed in Java?

17. Differentiate between array and vector.

18. Why thread synchronization is important in multithreading process?

Handwritten notes:
27
13
20
5 12 = 10
6
16